

are defined, and their convolution sum representation in the time domain is derived. The concepts of causality and stability of LTI systems are introduced. Also discussed is an important class of LTI systems characterized by an input-output relation composed of a linear constant coefficient difference equation and the procedure for computing its output for a given input and initial conditions. Several types of classification of LTI discrete-time systems are then considered, of which the most common one is in terms of the length of the impulse response. The concept of the frequency response of an LTI discrete-time system is then introduced followed by a careful examination of the difference between phase and group delays associated with the frequency response.

4.11 Problems

4.1 Show that the down-sampler defined by Eq. (2.24) is a time-varying system.

4.2 Show that the discrete-time systems described by the following equations are linear systems:

(a) Eq. (2.19), (b) Eq. (2.23), (c) Eq. (2.24), (d) Eq. (4.4), (e) Eq. (4.8), and (f) Eq. (4.9).

HW2 ✓ 4.3 For each of the following discrete-time systems, where $y[n]$ and $x[n]$ are, respectively, the output and the input sequences, determine whether or not the system is (1) linear, (2) causal, (3) stable, and (4) shift-invariant:

(a) $y[n] = x[n + 3]$;

(b) $y[n] = x[2 - n] + \alpha$, where α is a nonzero constant;

(c) $y[n] = \ln(1 - |x[n]|)$;

(d) $y[n] = \beta + \sum_{\ell=-1}^3 x[n - \ell]$, where β is a nonzero constant.

4.4 The short-time energy of a discrete-time sequence with varying amplitudes with time provides a reasonable measure of the amplitude variations [Rab78]. The computation of the short-time energy involves the squaring operator given by

$$y[n] = (x[n])^2, \quad (4.107)$$

where $x[n]$ and $y[n]$ are the input and output signals. Is the squaring operator linear? Is it time-invariant? Is it causal?

4.5 The squaring operator defined in Eq. (4.107) is quite sensitive to large amplitudes of the signal. To get around this problem, an average magnitude function is sometimes used to provide a measure of the amplitude variations [Rab78]. The computation of the average magnitude function involves the computation of the absolute value of the signal sample:

$$y[n] = |x[n]|, \quad (4.108)$$

where $x[n]$ and $y[n]$ are the input and output signals. Is the absolute value operator linear? Is it time-invariant? Is it causal?

HW2 ✓ 4.6 Consider the following discrete-time systems: (a) $y[n] = n^K x[n]$, (b) $y[n] = x[n^K]$, where $x[n]$ and $y[n]$ are, respectively, the input and output of the system and K is an integer. Are the above systems linear or nonlinear?

✓ 4.7 Show that the median filter defined by Eq. (4.10) is a time-invariant system.

HW2 ✓ 4.8 The second derivative $y[n]$ of a sequence $x[n]$ at time instant n is usually approximated by

$$y[n] = x[n + 1] - 2x[n] + x[n - 1].$$

If $y[n]$ and $x[n]$ denote the output and input of a discrete-time system, is the system linear? Is it time-invariant? Is it causal?

4.9 A discrete-time system is characterized by [Kai80]

$$y[n] = x^2[n] - x[n-1]x[n+1],$$

where $y[n]$ and $x[n]$ are the output and input sequences. Is the above system linear? Is it time-invariant? Is it causal?

4.10 Consider a causal discrete-time system characterized by a first-order linear, constant-coefficient difference equation given by

$$y[n] = ay[n-1] + bx[n], \quad n \geq 0,$$

where $y[n]$ and $x[n]$ are, respectively, the output and input sequences. Compute the expression for the output sample $y[n]$ in terms of the initial condition $y[-1]$ and the input samples.

- Is the system time-invariant if $y[-1] = 1$? Is the system linear if $y[-1] = 1$?
- Repeat part (a) if $y[-1] = 0$.
- Generalize the results of parts (a) and (b) to the case of an N th-order causal discrete-time system given by Eq. (4.32).

4.11 Consider the discrete-time system characterized by the input-output relation [Cad87]

$$y[n] = \frac{1}{2} \left(y[n-1] + \frac{x[n]}{y[n-1]} \right), \quad (4.109)$$

where $x[n]$ and $y[n]$ are, respectively, the input and output sequences. Show that the output $y[n]$ of the above system for an input $x[n] = \alpha\mu[n]$ with $y[-1] = 1$ converges to $\sqrt{\alpha}$ as $n \rightarrow \infty$ where α is a positive number. Is the above system linear or nonlinear? Is it time-invariant? Justify your answer.

4.12 An algorithm for the calculation of the square root of a number α is given by [Mik92]

$$y[n] = x[n] - y^2[n-1] + y[n-1], \quad (4.110)$$

where $x[n] = \alpha\mu[n]$ with $0 < \alpha < 1$. If $x[n]$ and $y[n]$ are considered as the input and output of a discrete-time system, is the system linear or nonlinear? Is it time-invariant? As $n \rightarrow \infty$, show that $y[n] \rightarrow \sqrt{\alpha}$. Note that $y[-1]$ is a suitable initial approximation to $\sqrt{\alpha}$.

4.13 Determine the expression for the impulse response of the factor-of-3 linear interpolator of Eq. (4.9).

4.14 Determine the expression for the impulse response of the factor-of- L linear interpolator.

4.15 A causal LTI discrete-time system is said to have an *overshoot* in its step response if the response exhibits an oscillatory behavior with decaying amplitudes around a final constant value. Show that the system has no overshoot in its step response if the impulse response $h[n]$ of the system is nonnegative for all $n \geq 0$.

4.16 Prove that the BIBO stability condition of Eq. (4.20) also holds for an LTI digital filter with a complex impulse response.

4.17 Prove that the convolution sum operation is commutative and distributive.

4.18 Consider the following three sequences:

$$x_1[n] = A \text{ (a constant)}, \quad x_2[n] = \mu[n], \quad x_3[n] = \begin{cases} 1, & \text{for } n = 0, \\ -1, & \text{for } n = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Show that $x_3[n] \otimes x_2[n] \otimes x_1[n] \neq x_2[n] \otimes x_3[n] \otimes x_1[n]$.

- 4.19** Prove that the convolution operation is associative for stable and single-sided sequences.
- 4.20** Develop closed-form expressions for the following convolution sums: (a) $\alpha^n \mu[n] \otimes \mu[n]$, (b) $n\alpha^n \mu[n] \otimes \mu[n]$.
- 4.21** Develop a general expression for the output $y[n]$ of an LTI discrete-time system in terms of its input $x[n]$ and the unit step response $s[n]$ of the system.
- 4.22** A periodic sequence $\tilde{x}[n]$ with a period N is applied as an input to an LTI discrete-time system characterized by an impulse response $h[n]$ generating an output $y[n]$. Is $y[n]$ a periodic sequence? If it is, what is its period?
- 4.23** Let $y[n]$ be the sequence obtained by a linear convolution of two causal finite-length sequences $h[n]$ and $x[n]$. For each pair of $y[n]$ and $h[n]$ listed below, determine $x[n]$. The first sample in each sequence is at time index $n = 0$.
- (a) $\{y[n]\} = \{6, -7, -13, 24, 12, 33, -5, 14\}$, $\{h[n]\} = \{3, 4, -1, 2\}$,
 - (b) $\{y[n]\} = \{-6, 17, 3, -38, 6, 41, -3, -20\}$, $\{h[n]\} = \{-2, 3, 1, 4\}$,
 - (c) $\{y[n]\} = \{15, -4, -16, 24, 36, -12, -31, -12\}$, $\{h[n]\} = \{3, -2, 0, 5, 4\}$.
- 4.24** An LTI discrete-time system is characterized by a left-side impulse response given by $h[n] = \alpha^n \mu[-n - 1]$. Determine the range of the value of the constant α for which the system is BIBO stable.
- 4.25** Prove Eq. (4.28).
- 4.26** Prove Eq. (4.31).
- 4.27** Consider a cascade of two causal stable LTI systems characterized by impulse responses $\alpha^n \mu[n]$ and $\beta^n \mu[n]$, where $0 < \alpha < 1$ and $0 < \beta < 1$. Determine the expression for the impulse response $h[n]$ of the cascade.
- 4.28** Determine the impulse response $g[n]$ of the inverse system of the LTI discrete-time system of Example 4.16.
- 4.29** Determine the impulse response $g[n]$ characterizing the inverse system of the cascaded LTI discrete-time system of Problem 4.27.
- 4.30** Determine the expression for the impulse response of each of the LTI systems shown in Figure P4.1.

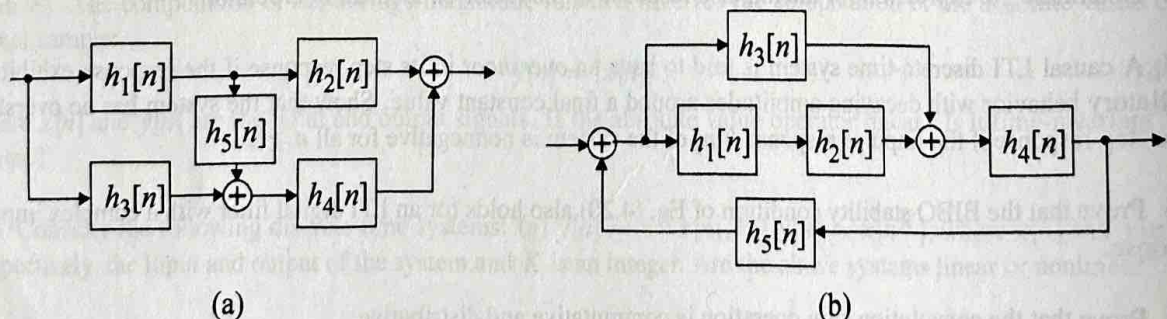


Figure P4.1

- 4.31** Determine the overall impulse response of the system of Figure P4.2, where the impulse responses of the component systems are: $h_1[n] = 2\delta[n - 2] + 3\delta[n + 1]$, $h_2[n] = \delta[n - 1] - 2\delta[n + 2]$, and $h_3[n] = 5\delta[n - 5] - 7\delta[n - 3] + 2\delta[n - 1] + \delta[n] - 3\delta[n + 1]$.

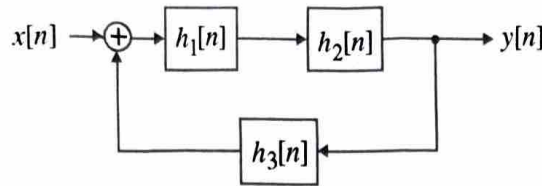


Figure P4.2

✓ 4.32 Is the cascade connection of two stable LTI systems also stable? Justify your answer.

✓ 4.33 Prove that the cascade connection of two passive (lossless) LTI systems is also passive (lossless).

✓ 4.34 Is the parallel connection of two stable LTI systems also stable? Justify your answer.

✓ 4.35 Is the parallel connection of two passive (lossless) LTI systems also passive (lossless)? Justify your answer.

4.36 Consider the causal LTI system described by the difference equation

$$y[n] = p_0 x[n] + p_1 x[n-1] - d_1 y[n-1],$$

where $x[n]$ and $y[n]$ denote, respectively, its input and output. Determine the difference equation representation of its inverse system.

4.37 Derive Eq. (4.53) by induction by first evaluating Eq. (4.50) for $n = 0, 1, 2, \dots$, and then solving for $h[0]$, $h[1]$, $h[2]$, etc.

4.38 Show by back substitution that the set of equations given by Eq. (4.55) reduces to Eq. (4.33).

✓ 4.39 The sequence of Fibonacci numbers $f[n]$ is a causal sequence defined by

$$f[n] = f[n-1] + f[n-2], \quad n \geq 2,$$

with $f[0] = 0$ and $f[1] = 1$.

(a) Develop an exact formula to calculate $f[n]$ directly for any n .

(b) Show that $f[n]$ is the impulse response of a causal LTI system described by the difference equation [Joh89]

$$y[n] = y[n-1] + y[n-2] + x[n-1].$$

4.40 Consider a first-order complex digital filter characterized by a difference equation

$$y[n] = \alpha y[n-1] + x[n],$$

where $x[n]$ is the real input sequence, $y[n] = y_{re}[n] + jy_{im}[n]$ is the complex output sequence with $y_{re}[n]$ and $y_{im}[n]$ denoting its real and imaginary parts, and $\alpha = a + jb$ is a complex constant. Develop an equivalent two-output, single-input real difference equation representation of the above complex digital filter. Show that the single-input, single-output digital filter relating $y_{re}[n]$ to $x[n]$ is described by a second-order difference equation.

4.41 Let $h[n]$, $h[n+1]$, and $h[n+2]$ denote the three consecutive impulse response samples of the first-order causal LTI system of Problem 4.36. Show that the coefficients of the difference equation, p_0 , p_1 , and d_1 , characterizing this system can be uniquely determined from the above three impulse response samples.

- ✓ 4.42 Determine the total solution for $n \geq 0$ of the difference equation

$$y[n] - 0.16y[n-1] = 5.88\mu[n],$$

with the initial condition $y[-1] = 5$.

- ✓ 4.43 Determine the total solution for $n \geq 0$ of the difference equation

$$y[n] - 0.7y[n-1] - 0.02y[n-2] = 3^n\mu[n],$$

with the initial condition $y[-1] = 3$, and $y[-2] = 0$.

- ✓ 4.44 Determine the total solution for $n \geq 0$ of the difference equation

$$y[n] - 0.3y[n-1] - 0.04y[n-2] = x[n] + 2x[n-1],$$

with the initial condition $y[-1] = 3$, and $y[-2] = 0$, when the forcing function is $x[n] = 3^n\mu[n]$.

- ✓ 4.45 Determine the impulse response $h[n]$ of the LTI system of Problem 4.42.

- ✓ 4.46 Determine the impulse response $h[n]$ of the LTI system of Problem 4.44.

- 4.47 Determine the step response of an LTI discrete-time system characterized by an impulse response $h[n] = (-\alpha)^n\mu[n]$, $0 < \alpha < 1$.

- 4.48 Show that the sum $\sum_{n=0}^{\infty} |n^K(\lambda_i)^n|$ converges if $|\lambda_i| < 1$.

- 4.49 Let a causal IIR digital filter be described by the difference equation of Eq. (4.32) where $y[n]$ and $x[n]$ denote the output and the input sequences, respectively. If $h[n]$ denotes its impulse response, show that

$$p_k = \sum_{n=0}^k h[n]d_{k-n}, \quad k = 0, 1, \dots, M.$$

From the above result, show that $p_n = h[n] \otimes d_n$.

- 4.50 Consider a causal FIR filter of length $L+1$ with an impulse response given by $\{g[n]\}$, $n = 0, 1, \dots, L$. Develop the difference equation representation of the form of Eq. (4.32) where $M+N=L$ of a causal finite-dimensional IIR digital filter with an impulse response $\{h[n]\}$ such that $h[n] = g[n]$ for $n = 0, 1, \dots, L$.

- 4.51 Compute the output of the accumulator of Eq. (4.2) for an input $x[n] = n\mu[n]$ and the following initial conditions: (a) $y[-1] = 1$, (b) $y[-1] = -1$, and (c) $y[-1] = 0$.

- 4.52 In the rectangular method of numerical integration, the integral on the right-hand side of Eq. (4.61) is expressed as

$$\int_{(n-1)T}^{nT} x(\tau)d\tau = T \cdot x((n-1)T). \quad (4.111)$$

Develop the difference equation representation of the rectangular method of numerical integration.

- 4.53 Develop a recursive implementation of the time-varying linear discrete-time system characterized by

$$y[n] = \begin{cases} \frac{1}{n} \sum_{\ell=1}^n x[\ell], & n > 0, \\ 0, & n \leq 0. \end{cases}$$

✓ 4.54 Show that the sequence $u[n] = z^n$, where z is a complex constant, is an eigenfunction of an LTI discrete-time system. Is the sequence $v[n] = z^n \mu[n]$ with z a complex constant also an eigenfunction of an LTI discrete-time system? Justify your answer.

4.55 Determine a closed-form expression for the frequency response $H(e^{j\omega})$ of the LTI discrete-time system characterized by an impulse response

$$h[n] = \delta[n] - \alpha\delta[n - R], \quad (4.112)$$

where $|\alpha| < 1$. What are the maximum and the minimum values of its magnitude response? How many peaks and dips of the magnitude response occur in the range $0 \leq \omega < 2\pi$? What are the locations of the peaks and the dips? Sketch the magnitude and the phase responses for $R = 5$.

✓ 4.56 Determine a closed-form expression for the frequency response $G(e^{j\omega})$ of the LTI discrete-time system characterized by an impulse response

$$g[n] = h[n] \otimes h[n] \otimes h[n], \quad (4.113)$$

where $h[n]$ is given by Eq. (4.112).

✓ 4.57 Determine a closed-form expression for the frequency response $G(e^{j\omega})$ of an LTI discrete-time system with an impulse response given by

$$g[n] = \begin{cases} \alpha^n, & 0 \leq n \leq M-1, \\ 0 & \text{otherwise,} \end{cases}$$

where $|\alpha| < 1$. What is the relation of $G(e^{j\omega})$ to $H(e^{j\omega})$ of Eq. (4.82)? Scale the impulse response by multiplying it with a suitable constant so that the dc value of the magnitude response is unity.

4.58 Determine the input-output relation of a factor-of-2 up-sampler in the frequency domain.

4.59 Determine and plot the DTFT of the impulse response of the cascade of the LTI discrete-time systems with two-sided impulse responses given by $h_1[n] = \delta[n] - \frac{\sin \omega_1 n}{\pi n}$, and $h_2[n] = \frac{\sin \omega_2 n}{\pi n}$, respectively, where $0 < \omega_1 < \omega_2 < \pi$.

✓ 4.60 Determine the frequencies present in the output of each of the following discrete-time systems for an input $x[n] = \cos(\omega_0 n)$:

$$(a) y_a[n] = \cos(\pi n/5)x[n], \quad (b) y_b[n] = x^4[n], \quad (c) y_c[n] = x[4n].$$

✓ 4.61 A noncausal LTI FIR discrete-time system is characterized by an impulse response

$$h[n] = a_1\delta[n-3] + a_2\delta[n-2] + a_3\delta[n-1] + a_4\delta[n] + a_5\delta[n+1] + a_6\delta[n+2].$$

For what values of the impulse response samples $\{a_i\}$, $1 \leq i \leq 6$, will its frequency response $H(e^{j\omega})$ have a zero phase?

4.62 A causal LTI FIR discrete-time system is characterized by an impulse response

$$h[n] = a_1\delta[n] + a_2\delta[n-1] + a_3\delta[n-2] + a_4\delta[n-3] + a_5\delta[n-4] + a_6\delta[n-5] + a_7\delta[n-6] + a_8\delta[n-7].$$

For what values of the impulse response samples $\{a_i\}$, $1 \leq i \leq 8$, will its frequency response $H(e^{j\omega})$ have a linear phase?

4.63 An FIR LTI discrete-time system is described by the difference equation

$$y[n] = a_1x[n+k] + a_2x[n+k-1] + a_3x[n+k-2] + a_2x[n+k-3] + a_1x[n+k-4],$$

where $y[n]$ and $x[n]$ denote, respectively, the output and the input sequences. Determine the expression for its frequency response $H(e^{j\omega})$. For what values of the constant k will the system have a frequency response $H(e^{j\omega})$ that is a real function of ω ?

✓ 4.64 Consider the cascade of two causal LTI systems: $h_1[n] = \alpha\delta[n] + \delta[n-1]$, and $h_2[n] = \beta^n\mu[n]$, $|\beta| < 1$. Determine the frequency response $H(e^{j\omega})$ of the overall system. For what values of α and β will $|H(e^{j\omega})| = 1$?

4.65 The input-output relation of a nonlinear discrete-time system in the frequency domain is given by

$$Y(e^{j\omega}) = |X(e^{j\omega})|^\alpha e^{j\arg X(e^{j\omega})}, \quad (4.114)$$

where $0 < \alpha \leq 1$, and $X(e^{j\omega})$ and $Y(e^{j\omega})$ denote the DTFTs of the input and output sequences. Determine the expression for its frequency response $H(e^{j\omega}) = Y(e^{j\omega})/X(e^{j\omega})$ and show that it has zero phase. The nonlinear algorithm described by Eq. (4.114) is known as the *alpha-rooting method* and has been used in image enhancement [Jai89].

4.66 Determine the expression for the frequency response $H(e^{j\omega})$ of a causal IIR LTI discrete-time system characterized by the input-output relation

$$y[n] = x[n] + \alpha y[n-R], \quad |\alpha| < 1,$$

where $y[n]$ and $x[n]$ denote, respectively, the output and the input sequences. Determine the maximum and the minimum values of its magnitude response. How many peaks and dips of the magnitude response occur in the range $0 \leq \omega < 2\pi$? What are the locations of the peaks and the dips? Sketch the magnitude and the phase responses for $R = 5$.

4.67 An IIR LTI discrete-time system is described by the difference equation

$$y[n] + a_1 y[n-1] + a_2 y[n-2] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2],$$

where $y[n]$ and $x[n]$ denote, respectively, the output and the input sequences. Determine the expression for its frequency response. For what values of the constants b_i will the magnitude response be a constant for all values of ω ?

✓ 4.68 Consider an LTI discrete-time system with an impulse response $h[n] = (0.4)^n \mu[n]$. Determine the frequency response $H(e^{j\omega})$ of the system and evaluate its value at $\omega = \pm\pi/4$. What is the steady-state output $y[n]$ of the system for an input $x[n] = \sin(\pi n/4)\mu[n]$?

4.69 An FIR filter of length 3 is defined by a symmetric impulse response, i.e., $h[0] = h[2]$. Let the input to this filter be a sum of two cosine sequences of angular frequencies 0.2 rad/samples and 0.5 rad/samples, respectively. Determine the impulse response coefficients so that the filter passes only the high-frequency component of the input.

4.70 (a) Design a length-5 FIR bandpass filter with an antisymmetric impulse response $h[n]$, i.e., $h[n] = -h[4-n]$, $0 \leq n \leq 4$, satisfying the following magnitude response values: $|H(e^{j\pi/4})| = 0.5$ and $|H(e^{j\pi/2})| = 1$.

(b) Determine the exact expression for the frequency response of the filter designed, and plot its magnitude and phase responses.

4.71 (a) Design a length-4 FIR bandpass filter with an antisymmetric impulse response $h[n]$, i.e., $h[n] = -h[3-n]$, $0 \leq n \leq 3$, satisfying the following magnitude response values: $|H(e^{j\pi/4})| = 1$ and $|H(e^{j\pi/2})| = 0.5$.

(b) Determine the exact expression for the frequency response of the filter designed, and plot its magnitude and phase responses.

4.72 Consider an FIR filter of length 5 with a symmetric impulse response, i.e., $h[n] = h[4-n]$, $0 \leq n \leq 4$. An input consisting of a sum of three cosine sequences of angular frequencies: 0.3 rad/samples, 0.4 rad/samples, and 0.7 rad/samples, respectively, is applied to this filter. Determine the impulse response coefficients so that the filter passes only the midfrequency component of the input.

4.73 Consider the two LTI causal digital filters with impulse responses given by

$$h_A[n] = 0.3\delta[n] - \delta[n-1] + 0.3\delta[n-2], \quad h_B[n] = 0.3\delta[n] + \delta[n-1] + 0.3\delta[n-2].$$

- (a) Sketch the magnitude responses of the two filters and compare their characteristics.
 (b) Let $h_A[n]$ be the impulse response of a causal digital filter with a frequency response $H_A(e^{j\omega})$. Define another digital filter whose impulse response $h_C[n]$ is given by

$$h_C[n] = (-1)^n h_A[n], \quad \text{for all } n.$$

What is the relation between the frequency response $H_C(e^{j\omega})$ of this new filter and the frequency response $H_A(e^{j\omega})$ of the parent filter?

4.74 As indicated in Example 4.30, the trapezoidal integration formula can be represented as an IIR digital filter represented by a difference equation given by

$$y[n] = y[n-1] + \frac{1}{2}\{x[n] + x[n-1]\},$$

with $y[-1] = 0$. Determine the frequency response of the above filter.

4.75 A recursive difference equation representation of the Simpson's numerical integration formula is given by [Ham89]

$$y[n] = y[n-2] + \frac{1}{3}\{x[n] + 4x[n-1] + x[n-2]\}.$$

Evaluate the frequency response of the above filter and compare it with that of the trapezoidal method of Problem 4.74.

4.76 An LTI causal discrete-time system is characterized by the difference equation

$$y[n] = d_3x[n] + d_2x[n-1] + d_1x[n-2] + x[n-3] - d_1y[n-1] - d_2y[n-2] - d_3y[n-3],$$

where $y[n]$ and $x[n]$ are, respectively, the output and input sequences. Determine the expression for the frequency response of the system and show that it has a unity magnitude response for all values of ω .

4.77 Show that the group delay $\tau(\omega)$ of an LTI discrete-time system characterized by a frequency response $H(e^{j\omega})$ can be expressed as

$$\tau(\omega) = \text{Re} \left\{ j \frac{dH(e^{j\omega})}{d\omega} \frac{1}{H(e^{j\omega})} \right\}. \quad (4.115)$$

4.78 The frequency response of an LTI FIR discrete-time system is given by $H(e^{j\omega}) = \alpha_0 + \alpha_1 e^{-j\omega} + \alpha_2 e^{-j2\omega} + \alpha_3 e^{-j3\omega} + \alpha_4 e^{-j4\omega}$. For what relations between the coefficients α_i , $0 \leq i \leq 4$ will $H(e^{j\omega})$ have a constant group delay?

4.79 Determine the expressions for the group delay of each of the LTI systems whose frequency responses are given below:

(a) $H_a(e^{j\omega}) = \alpha + \beta e^{-j\omega},$

(b) $H_b(e^{j\omega}) = \frac{1}{1+\gamma e^{-j\omega}}, |\gamma| < 1,$

(c) $H_c(e^{j\omega}) = \frac{\alpha+\beta e^{-j\omega}}{1+\gamma e^{-j\omega}}, |\gamma| < 1,$

(d) $H_d(e^{j\omega}) = \frac{1}{(1+\gamma e^{-j\omega})(1+\delta e^{-j\omega})}, |\gamma| < 1, |\delta| < 1,$

(e) $H_e(e^{j\omega}) = \frac{\alpha+\beta e^{-j\omega}}{(1+\gamma e^{-j\omega})(1+\delta e^{-j\omega})}, |\gamma| < 1, |\delta| < 1.$

4.80 Let $H(e^{j\omega})$ denote the frequency response of an LTI discrete-time system with an impulse response $h[n]$ and let $G(e^{j\omega})$ denote the Fourier transform of the sequence $nh[n]$. Show that the group delay of the LTI system can be computed using

$$\tau_g(\omega) = \frac{H_{\text{re}}(e^{j\omega})G_{\text{re}}(e^{j\omega}) + H_{\text{im}}(e^{j\omega})G_{\text{im}}(e^{j\omega})}{|H(e^{j\omega})|^2}, \quad (4.116)$$

where $H_{\text{re}}(e^{j\omega})$ and $H_{\text{im}}(e^{j\omega})$ denote the real and imaginary parts of $H(e^{j\omega})$, respectively, and $G_{\text{re}}(e^{j\omega})$ and $G_{\text{im}}(e^{j\omega})$ denote the real and imaginary parts of $G(e^{j\omega})$, respectively.

4.81 Using Eq. (4.116), determine the group delays of the LTI discrete-time systems with frequency responses as given below:

✓ (a) $H_a(e^{j\omega}) = 2 - 0.5e^{-j\omega}$,

(b) $H_b(e^{j\omega}) = \frac{1}{1+0.8e^{-j\omega}}$,

(c) $H_c(e^{j\omega}) = \frac{0.3+0.7e^{-j\omega}}{1+0.5e^{-j\omega}}$,

(d) $H_d(e^{j\omega}) = \frac{1}{(0.3+0.4e^{-j\omega})(1+0.5e^{-j\omega})}$,

(e) $H_e(e^{j\omega}) = \frac{2-0.5e^{-j\omega}}{(1-0.4e^{-j\omega})(1-0.6e^{-j\omega})}$.

4.12 MATLAB Exercises

M 4.1 Using Program 4_1, investigate the effect of signal smoothing by a moving-average filter of lengths 5, 7, and 9. Does the signal smoothing improve with an increase in the length? What is the effect of the length on the delay between the smoothed output and the noisy input?

M 4.2 Write a MATLAB program implementing the discrete-time system of Eq. (4.109) in Problem 4.11, and show that the output $y[n]$ of this system for an input $x[n] = \alpha\mu[n]$ with $y[-1] = 1$ converges to $\sqrt{\alpha}$ as $n \rightarrow \infty$.

M 4.3 Write a MATLAB program to compute the square root using the algorithm of Eq. (4.110) in Problem 4.12, and show that the output $y[n]$ of this system for an input $x[n] = \alpha\mu[n]$ with $y[-1] = 1$ converges to $\sqrt{\alpha}$ as $n \rightarrow \infty$. Plot the error as a function of n for several different values of α . How would you compute the square root of a number α with a value greater than one?

M 4.4 Write a MATLAB program to compute the group delay using the expression of Problem 4.77 at a prescribed set of discrete frequencies.

M 4.5 Write a MATLAB program to simulate the filter designed in Problem 4.69, and verify its filtering operation.

M 4.6 Write a MATLAB program to simulate the filter designed in Problem 4.72, and verify its filtering operation.